WORKED SOLUTION

STEP III 1989, Question 12

Mechanics • Circular Motion

STEP III 1989, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q12

The key idea

The plane is smooth and the rod is light and freely pivoted, so the only horizontal force on P acts along the rod. The acceleration of P perpendicular to AP must therefore be zero. Once we have worked out that component, setting it to zero gives a pendulum equation for ϕ, and the usual "multiply by $\dot{\phi}$ and integrate" trick gives $\dot{\phi}$.

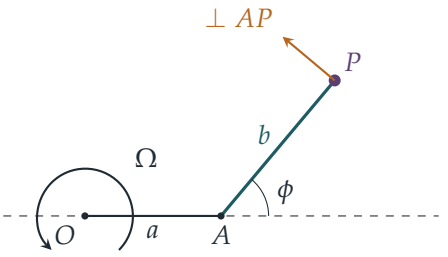
Solution

Model and set-up

Model P as a particle on a smooth plane and the rod as light and rigid, freely pivoted at A. Work in a fixed (inertial) frame. Use complex numbers for positions in the plane, with O at the origin and the real axis along OA at $t = 0$, and take $\Omega > 0$ anticlockwise. Since A is fixed in the plane, it moves on a circle of radius a, and AP makes angle ϕ with OA produced, which itself has turned through Ωt. So, writing $\theta = \Omega t + \phi$,

$$z_A = a\,\mathrm{e}^{\mathrm{i}\Omega t}, \qquad z_P = a\,\mathrm{e}^{\mathrm{i}\Omega t} + b\,\mathrm{e}^{\mathrm{i}\theta}.$$

Here $\mathrm{e}^{\mathrm{i}\theta}$ is the unit vector along AP, and $\mathrm{i}\,\mathrm{e}^{\mathrm{i}\theta}$ is the unit vector perpendicular to AP in the direction of increasing ϕ.



The perpendicular component of acceleration

Differentiating, with $\dot\theta = \Omega + \dot\phi$ and $\ddot\theta = \ddot\phi$ (as Ω is constant),

$$\dot z_P = \mathrm{i}a\Omega\,\mathrm{e}^{\mathrm{i}\Omega t} + \mathrm{i}b(\Omega+\dot\phi)\,\mathrm{e}^{\mathrm{i}\theta},$$
$$\ddot z_P = -a\Omega^2\,\mathrm{e}^{\mathrm{i}\Omega t} + b(\mathrm{i}\ddot\phi - (\Omega+\dot\phi)^2)\mathrm{e}^{\mathrm{i}\theta}.$$

To resolve along and perpendicular to AP, write $\mathrm{e}^{\mathrm{i}\Omega t} = \mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{-\mathrm{i}\phi} = \mathrm{e}^{\mathrm{i}\theta}(\cos\phi - \mathrm{i}\sin\phi)$. Then

$$\ddot z_P = [-a\Omega^2\cos\phi - b(\Omega+\dot\phi)^2]\mathrm{e}^{\mathrm{i}\theta} + [\,b\ddot\phi + a\Omega^2\sin\phi\,]\,\mathrm{i}\,\mathrm{e}^{\mathrm{i}\theta}.$$

So the component of the acceleration of P perpendicular to AP, in the direction of increasing ϕ, is

$$\boxed{b\,\frac{\mathrm{d}^2\phi}{\mathrm{d}t^2} + a\Omega^2\sin\phi}$$

(The first term is the familiar $b\ddot\phi$ of motion on a circle about A. The second comes from A's own centripetal acceleration $a\Omega^2$ towards O, resolved perpendicular to the rod.)

Finding $\mathrm{d}\phi/\mathrm{d}t$

Ask yourself Why must this component of acceleration be zero?

The forces on P are its weight and the normal reaction (both vertical, and they balance), no friction (the plane is smooth), and the force from the rod. A light rod that is freely pivoted at A can only push or pull along its own length. So there is no horizontal force perpendicular to AP, and Newton's second law gives

$$b\ddot\phi + a\Omega^2\sin\phi = 0.$$

This is the pendulum equation. Multiplying by $\dot\phi$ and integrating,

$$\tfrac12 b\dot\phi^2 = a\Omega^2\cos\phi + C.$$

Initial conditions. Initially O, A, P are in a straight line. Since $b > a$, P cannot lie between O and A, and we take P to be beyond A, so $\phi = 0$ at $t = 0$. Then both exponentials equal 1 and

$$\dot z_P(0) = \mathrm{i}[(a+b)\Omega + b\dot\phi(0)],$$

which is perpendicular to OP (the real axis). Note that $(a+b)\Omega$ is just the speed of the point of the plane underneath P, and $b\dot\phi(0)$ is P's speed relative to the plane. The printed speed is $(a+2\sqrt{ab})\Omega$, but this looks like a misprint for $(a+b+2\sqrt{ab})\Omega = (\sqrt{a}+\sqrt{b})^2\Omega$, which is the value that gives the stated result (see the remark at the end). With that value, P moves at $2\sqrt{ab}\,\Omega$ relative to the plane, so

$$\dot\phi(0) = 2\Omega\sqrt{a/b}.$$

Putting $\phi = 0$ into the energy equation gives $\tfrac12 b\cdot 4a\Omega^2/b = a\Omega^2 + C$, so $C = a\Omega^2$ and

$$\tfrac12 b\dot\phi^2 = a\Omega^2(1+\cos\phi) = 2a\Omega^2\cos^2\tfrac{\phi}{2}, \qquad \text{so} \qquad \dot\phi^2 = \frac{4a\Omega^2}{b}\cos^2\tfrac{\phi}{2}.$$

Initially $\dot\phi > 0$, and $\dot\phi$ can change sign only by passing through zero, which needs $\cos\frac{\phi}{2} = 0$, that is $\phi = \pm\pi$. So as long as $-\pi < \phi < \pi$ we take the positive root:

$$\boxed{\frac{\mathrm{d}\phi}{\mathrm{d}t} = 2\Omega\sqrt{\frac{a}{b}}\cos\frac{\phi}{2}}$$

P never crosses OA

? **Ask yourself** Couldn't ϕ just creep up to π and go past it?

P is on the line OA only when ϕ is a multiple of π (for $\phi = \pi$ it lies a distance $b - a$ beyond O). Since $\dot\phi > 0$ while $0 \le \phi < \pi$, ϕ increases from 0, and the only way back onto the line is to reach $\phi = \pi$. We show that this never happens by solving the equation. Separating the variables, with $k = \sqrt{a/b}$,

$$\int_0^{\phi} \sec\tfrac{u}{2}\,\mathrm{d}u = 2k\Omega t \quad\Longrightarrow\quad \ln\Big(\sec\tfrac{\phi}{2} + \tan\tfrac{\phi}{2}\Big) = k\Omega t.$$

With $s = \sin\frac{\phi}{2}$ we have $\sec\frac{\phi}{2} + \tan\frac{\phi}{2} = \frac{1+s}{\sqrt{1-s^2}} = \sqrt{\frac{1+s}{1-s}}$, so the left-hand side is $\frac{1}{2}\ln\frac{1+s}{1-s} = \operatorname{artanh} s$. Hence

$$\sin\frac{\phi}{2} = \tanh\Big(\Omega t\sqrt{\tfrac{a}{b}}\Big).$$

For every $t > 0$ the right-hand side lies strictly between 0 and 1, so $0 < \phi < \pi$ for all $t > 0$. Then P always stays strictly on one side of the line OA. It approaches the line (at the point beyond O) as $t \to \infty$ but never reaches it, so P never crosses OA. ■

(Without solving: $\phi = \pi$, $\dot\phi = 0$ is an equilibrium of the pendulum equation, and near it $\psi = \pi - \phi$ satisfies $\dot\psi \approx -k\Omega\psi$, so ψ decays exponentially and is never zero.)

Remark on the printed speed. If we take the speed literally as $(a + 2\sqrt{ab})\Omega$, then $\dot\phi(0) = (2k-1)\Omega$, and the same integration gives $\dot\phi^2 = \Omega^2[(2k-1)^2 - 4k^2\sin^2\frac{\phi}{2}]$. This is not a neat expression. P then oscillates about $\phi = 0$ (crossing OA produced every swing) if $b < 16a$, and goes right round, crossing OA, if $b > 16a$. So the printed value cannot be what was intended.

Checks

- Setting $a = 0$ puts A at O, and then $\dot\phi \equiv 0$: P simply rides round with the plane at speed $b\Omega$. That is right, because the initial speed becomes $b\Omega$.
- Units: $\Omega\sqrt{a/b}$ is a rate, so it has units of s^{-1}, as $\dot\phi$ should.
- We stepped $b\ddot\phi = -a\Omega^2\sin\phi$ numerically (RK4) with $a = 1$, $b = 3$, $\Omega = 1.3$. At $t = 2$ it gave $\phi \approx 2.2644$ and $\dot\phi \approx 0.6375$, agreeing with $2\arcsin\tanh(\Omega t\sqrt{a/b})$ and $2\Omega\sqrt{a/b}\cos\frac{\phi}{2}$ to 10 significant figures. At $t = 10$, $\phi \approx 3.1394 < \pi$.

Common slips

- Forgetting the frame's own contribution $(a+b)\Omega$ to P's initial velocity, and treating the given speed as speed relative to the plane.
- Writing the acceleration of P as just that of circular motion about A. The pivot A is itself accelerating, which is where the $a\Omega^2 \sin\phi$ comes from.
- Getting the sign of $a\Omega^2 \sin\phi$ wrong. With the wrong sign, $\phi = 0$ would be unstable, which is unphysical because the "centrifugal" pull is away from O.

Beyond STEP

Viewed from the rotating plane, P is a pendulum hanging in a uniform "gravity" $a\Omega^2$ pointing along OA produced. The Coriolis force acts along the rod, so it does no work and drops out. The initial condition puts the motion exactly on the *separatrix*: the borderline between swinging and going over the top, where the pendulum takes infinite time to reach the inverted position. Rotating frames, fictitious forces and phase-plane pictures like this are covered in Kleppner and Kolenkow, *An Introduction to Mechanics*.

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